

CHAPTER 10

USING GATEWAY VPN

A virtual private network (VPN) is a network connection that establishes a secure tunnel between endpoints via a public source, such as the Internet or other network. You can configure the Sourcefire 3D System to build secure VPN tunnels between the virtual routers of Sourcefire managed devices. The system builds tunnels using the Internet Protocol Security (IPSec) protocol suite.

Only Sourcefire managed devices can be used as endpoints in Sourcefire VPN deployments. Third-party endpoints are not supported.

After the VPN connection is established, the hosts behind the local gateway can connect to the hosts behind the remote gateway through the secure VPN tunnel. A connection consists of the IP addresses and host names of the two gateways, the subnets behind them, and the shared secrets for the two gateways to authenticate to each other.

The VPN endpoints authenticate to each other with either the Internet Key Exchange (IKE) version 1 or version 2 protocol to create a security association for the tunnel. The system uses either the IPSec authentication header (AH) protocol or the IPSec encapsulating security payload (ESP) protocol to authenticate the data entering the tunnel. The ESP protocol encrypts the data as well as providing the same functionality as AH.

If you have access control policies in your deployment, the system does not send VPN traffic until it has passed through access control. In addition, the system does not send tunnel traffic to the public source when the tunnel is down.

To configure and apply VPN deployments, you must have a VPN license enabled on each of your target managed devices. Additionally, VPN features are only available on Series 3 devices.

See the following sections for more information on creating and managing VPN deployments:

- [Understanding IPSec](#) on page 333
- [Understanding VPN Deployments](#) on page 334
- [Managing VPN Deployments](#) on page 336

Understanding IPSec

The IPSec protocol suite defines how IP packets across a VPN tunnel are hashed, encrypted, and encapsulated in the ESP or AH security protocol. The Sourcefire 3D System uses the hash algorithm and encryption key of the Security Association (SA), which becomes established between the two gateways by the Internet Key Exchange (IKE) protocol.

Security associations (SA) establish shared security attributes between two devices and allow VPN endpoints to support secure communication. An SA allows two VPN endpoints to handle the parameters for how the VPN tunnel is secured between them.

The system uses the Internet Security Association and Key Management Protocol (ISAKMP) during the initial phase of negotiating the IPSec connection to establish the VPN between endpoints and the authenticated key exchange. The IKE protocol resides within ISAKMP. See [Understanding IKE](#) on page 333 for more information about the IKE protocol.

The AH security protocol provides protection for packet headers and data, but it cannot encrypt them. ESP provides encryption and protection for packets, but it cannot secure the outermost IP header. In many cases, this protection is not required, and most VPN deployments use ESP more frequently than AH because of its encryption capabilities. Since VPN only operates in tunnel mode, the system encrypts and authenticates the entire packet from Layer 3 and up in the ESP protocol. ESP in tunnel mode encrypts the data as well as providing the latter's encryption capabilities.

Understanding IKE

The Sourcefire 3D System uses the IKE protocol to mutually authenticate the two gateways against each other as well as to negotiate the SA for the tunnel. The process consists of two phases.

IKE phase 1 establishes a secure authenticated communication channel by using the Diffie-Hellman key exchange to generate a pre-shared key to encrypt further IKE communications. This negotiation results in a bidirectional ISAKMP security association. The system allows you to perform the authentication using a pre-shared key. Phase 1 operates in main mode, which seeks to protect all data during the negotiation, while also protecting the identity of the peers.

During IKE phase 2, the IKE peers use the secure channel established in phase 1 to negotiate security associations on behalf of IPSec. The negotiation results in a minimum of two unidirectional security associations, one inbound and one outbound.

Understanding VPN Deployments

A VPN deployment specifies the endpoints and networks that are included in a VPN and how they connect to each other. After you configure a VPN deployment, you can then apply it to your managed devices or devices managed by another Defense Center.

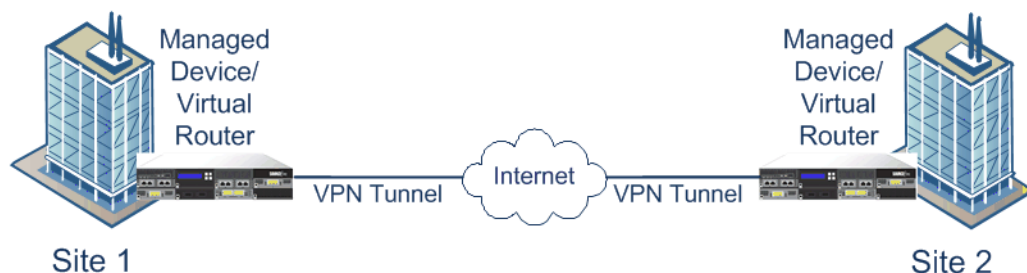
The system supports three types of VPN deployments: point-to-point, star, and mesh. See the following sections for more information about these VPN deployments:

- [Understanding Point-to-Point VPN Deployments](#) on page 334
- [Understanding Star VPN Deployments](#) on page 334
- [Understanding Mesh VPN Deployments](#) on page 335

Understanding Point-to-Point VPN Deployments

In a point-to-point VPN deployment, two endpoints communicate directly with each other. You configure the two endpoints as peer devices, and either device can initiate the secured connection. Each of the devices in this configuration must be a VPN-enabled managed device.

The following diagram displays a typical point-to-point VPN deployment.



See [Configuring Point-to-Point VPN Deployments](#) on page 338 for more information.

Understanding Star VPN Deployments

In a star VPN deployment, a central endpoint (hub node) establishes a secure connection with multiple remote endpoints (leaf nodes). Each connection between the hub node and an individual leaf node is a separate VPN tunnel. The

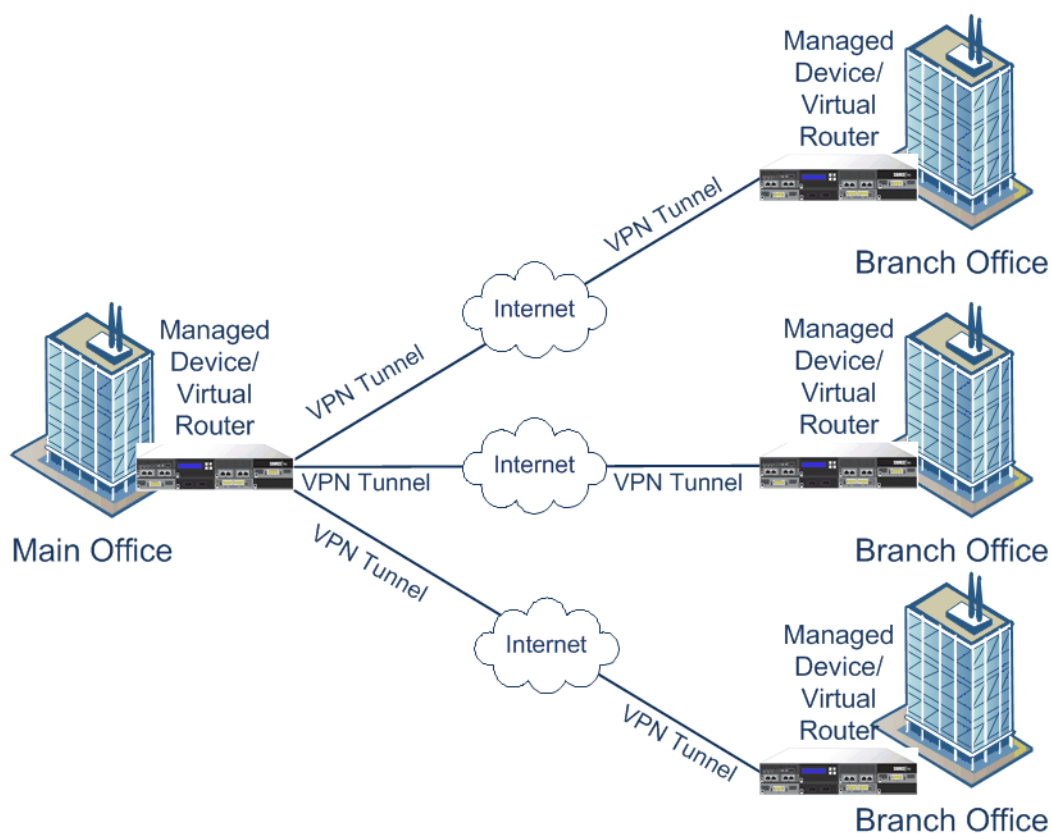
hosts behind any of the leaf nodes can communicate with each other through the hub node.

Star deployments commonly represent a VPN that connects an organization's main and branch office locations using secure connections over the Internet or other third-party network. Star VPN deployments provide all employees with controlled access to the organization's network.

In a typical star deployment, the hub node is located at the main office. Leaf nodes are located at branch offices and initiate most of the traffic. Each of the nodes must be a VPN-enabled managed device.

Note that star deployments only support IKE version 2.

The following diagram displays a typical star VPN deployment.



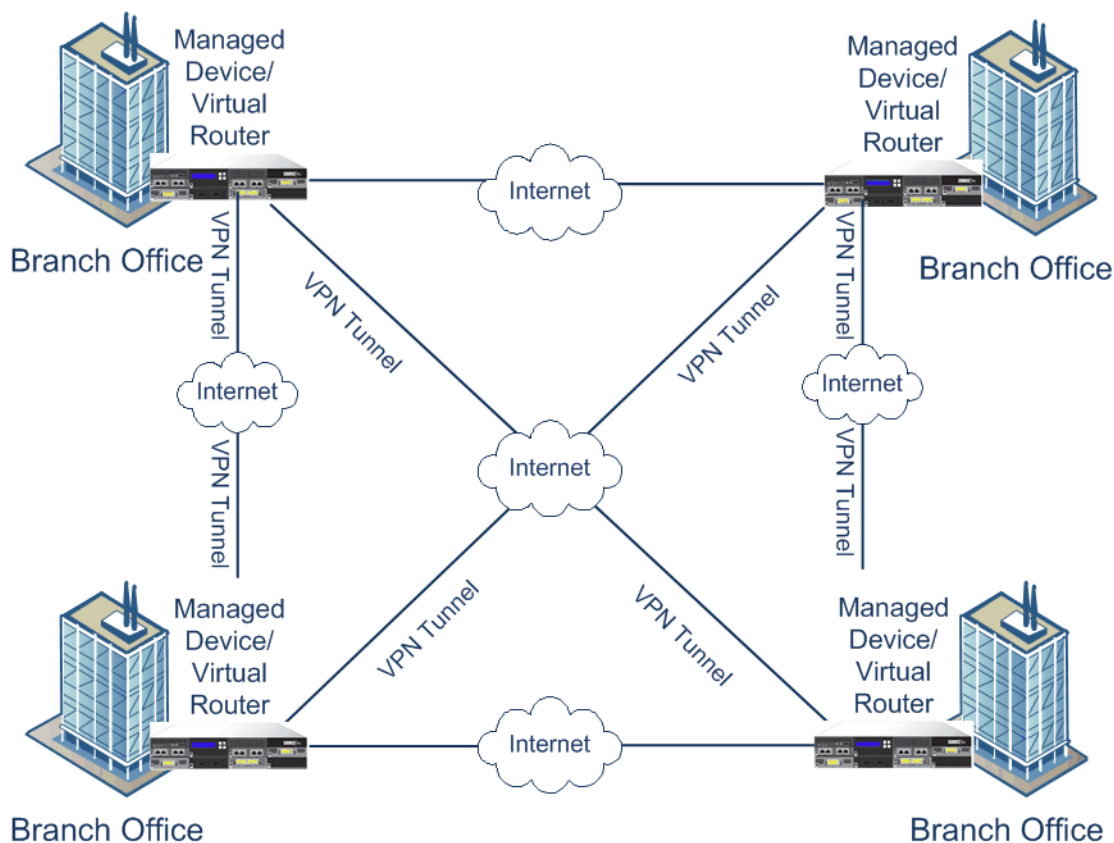
See [Configuring Star VPN Deployments](#) on page 342 for more information.

Understanding Mesh VPN Deployments

In a mesh VPN deployment, all endpoints can communicate with every other endpoint by means of an individual VPN tunnel. The mesh deployment offers redundancy so that when one endpoint fails, the remaining endpoints can still

communicate with each other. This type of deployment commonly represents a VPN that connects a group of decentralized branch office locations. The number of VPN-enabled managed devices you deploy in this configuration depends on the level of redundancy you require. Each of the endpoints must be a VPN-enabled managed device.

The following diagram displays a typical mesh VPN deployment.



See [Configuring Mesh VPN Deployments](#) on page 345 for more information.

Managing VPN Deployments

LICENSE: VPN

SUPPORTED DEVICES: Series 3

On the VPN page (**Devices > VPN**) you can view all of your current VPN deployments by name and the endpoints contained in the deployment. Options

on this page allow you to view the status of a VPN deployment, create a new deployment, apply a deployment, and edit or delete a deployment.

WARNING! If you select the default access control policy when registering a device to your Defense Center, the default access control rule blocks all traffic. If you configure a VPN deployment on the device, the deployment fails.

Note that when you register a device to a Defense Center, applied VPN deployments sync to the Defense Center during registration.

The following table describes the actions you can take to manage your deployments on the VPN page.

VPN Deployment Management Actions

To...	You CAN...
create a new VPN deployment	click Add . See Configuring VPN Deployments on page 337 for more information.
modify the settings in an existing VPN deployment	click the edit icon (✎). See Configuring VPN Deployments on page 337 for more information.
view the status of an existing VPN deployment	click the status icon. See Viewing VPN Deployment Status on page 351 for more information.
apply a VPN deployment to all devices targeted in the deployment	click the apply icon (✅). See Applying a VPN Deployment on page 351 for more information.
delete a VPN deployment	click the delete icon (🗑), then click Yes , or click No if you decide not to delete the deployment.

Configuring VPN Deployments

LICENSE: VPN

SUPPORTED DEVICES: Series 3

When you create a new VPN deployment you must, at minimum, give it a unique name, specify a deployment type, and designate a pre-shared key. You can select from three types of deployment, each containing a group of VPN tunnels:

- Point-to-point (PTP) deployments establish a VPN tunnel between two endpoints.
- Star deployments establish a group of VPN tunnels connecting a hub endpoint to a group of leaf endpoints.
- Mesh deployments establish a group of VPN tunnels among a set of endpoints.

Only Sourcefire managed devices can be used as endpoints in Sourcefire VPN deployments. Third-party endpoints are not supported.

You must define a pre-shared key for VPN authentication. You can specify a default key to use in all of the VPN connections you generate in a deployment. For point-to-point deployments, you can specify a pre-shared key for each endpoint pair.

See the following sections for more information on creating each type of VPN deployment:

- [Configuring Point-to-Point VPN Deployments](#) on page 338
- [Configuring Star VPN Deployments](#) on page 342
- [Configuring Mesh VPN Deployments](#) on page 345

Configuring Point-to-Point VPN Deployments

LICENSE: VPN

SUPPORTED DEVICES: Series 3

When configuring a point-to-point VPN deployment, you define a group of endpoint pairs and then create a VPN between the two nodes in each pair. For more information, see [Understanding Point-to-Point VPN Deployments](#) on page 334.

The following list describes the options you can specify in your deployment.

Name

Give the deployment a unique name.

Type

Click **PTP** to specify that you are configuring a point-to-point deployment.

Pre-shared Key

Define a unique pre-shared key for authentication. The system uses this key for all the VPNs in your deployment, unless you specify a pre-shared key for each endpoint pair.

Device

You can select a managed device, including a device stack or cluster, as an endpoint for your deployment. For Sourcefire managed devices not managed by the Defense Center you are using, select **Other** and then specify an IP address for the endpoint.

Virtual Router

If you selected a managed device as your endpoint, select a virtual router that is currently applied to the selected device. You cannot select the same virtual router for more than one endpoint.

Interface

If you selected a managed device as your endpoint, select a routed interface that is assigned to the selected virtual router.

IP Address

- If you selected a managed device as an endpoint, select an IP address that is assigned to the selected routed interface.
- If the managed device is a device cluster, you can only select from a list SFRP IP addresses.
- If you selected a managed device **not** managed by the Defense Center, specify an IP address for the endpoint.

Protected Networks

Specify the networks in your deployment that are encrypted. Enter a subnet with CIDR block for each network. IKE version 1 only supports a single protected network.

Note that VPN endpoints cannot have the same IP address and that protected networks in a VPN endpoint pair cannot overlap. If a list of protected networks for an endpoint contains one or more IPv4 or IPv6 entry, the other endpoint's protected network must have at least one entry of the same type (i.e., IPv4 or IPv6). If it does not, then the other endpoint's IP address must be of the same type and must not overlap with the entries in the protected network. (Use /32 CIDR address blocks for IPv4 and /128 CIDR address blocks for IPv6). If both of these checks fail, the endpoint pair is invalid.

Internal IP

Select the check box if the endpoint resides behind a firewall with network address translation.

Public IP

If you selected **Internal IP**, specify a public IP address for the firewall. If the endpoint is a responder, you must specify this value.

Public IKE Port

If you selected **Internal IP**, specify a single numerical value from 1 to 65535 for the UDP port on the firewall that is being port-forwarded to the internal endpoint. If the endpoint is a responder and the port on the firewall being forwarded is not 500 or 4500, you must specify this value.

Use Deployment Key

Select the check box to use the pre-shared key defined for the deployment. Clear the check box to specify a pre-shared key for VPN authentication for this endpoint pair.

Pre-shared Key

If you cleared the **Use Deployment Key** check box, specify a pre-shared key in this field.

TIP! To edit an existing point-to-point deployment, click the edit icon (✎) next to the deployment. You cannot edit the deployment type after you initially save the deployment. Two users should **not** edit the same deployment simultaneously; however, note that the web interface does not prevent simultaneous editing.

To configure a point-to-point VPN deployment:

ACCESS: Admin/Network Admin

1. Select **Devices > VPN**.

The VPN page appears.

2. Click **Add**.

The Create New VPN Deployment pop-up window appears.

Create New VPN Deployment

General Advanced

Name:

Type: ☒ PTP ☐ Star ☐ Mesh

Pre-shared Key:

Node Pairs:

Node A	Node B
--------	--------

Please enter a deployment name.

Save Cancel

3. Give the deployment a unique **Name**.
You can use all printable characters, including spaces and special characters.
4. Ensure that **PTP** is selected as the **Type**.
5. Give the deployment a unique **Pre-shared Key**.

6. Next to **Node Pairs**, click the add icon (+).
- The Add New Endpoint Pair pop-up window appears.

Add New Endpoint Pair

Node A:

Device:

Virtual Router:

Interface:

IP Address:

Protected Networks +

Network

Advanced:

Internal IP: ☐

Public IP:

public IKE Port:

Use Deployment Key: ☒

Node B:

Device:

Virtual Router:

Interface:

IP Address:

Protected Networks +

Network

Advanced:

Internal IP: ☐

Public IP:

public IKE Port:

Incomplete endpoint.

Save Cancel

7. Configure the VPN deployment, as described earlier in this section.
 8. Under **Node A**, next to **Protected Networks**, click the add icon (+).
- The Add Network pop-up window appears.
9. Type a CIDR block for the protected network.
 10. Click **OK**.
- The protected network is added.
11. Repeat step 8 through step 10 for **Node B**.
 12. Click **Save**.
- The endpoint pair is added to your deployment and the Create New VPN Deployment pop-up window appears again.
13. Click **Save** to finish configuring your deployment and the VPN page appears again.
- Note that you must apply the deployment for it to take effect; see [Applying a VPN Deployment](#) on page 351.

Configuring Star VPN Deployments

LICENSE: VPN

SUPPORTED DEVICES: Series 3

When configuring a star VPN deployment, you define a single hub node endpoint and a group of leaf node endpoints. You must define the hub node endpoint and at least one leaf node endpoint to configure the deployment. For more information, see [Understanding Star VPN Deployments](#) on page 334.

The following list describes the options you can specify in your deployment.

Name

Give the deployment a unique name.

Type

Click **Star** to specify that you are configuring a star deployment.

Pre-shared Key

Define a unique pre-shared key for authentication.

Device

You can select a managed device, including a device stack or cluster, as an endpoint for your deployment. For Sourcefire managed devices not managed by the Defense Center you are using, select **Other** and then specify an IP address for the endpoint.

Virtual Router

If you selected a managed device as your endpoint, select a virtual router that is currently applied to the selected device. You cannot select the same virtual router for more than one endpoint.

Interface

If you selected a managed device as your endpoint, select a routed interface that is assigned to the selected virtual router.

IP Address

- If you selected a managed device as an endpoint, select an IP address that is assigned to the selected routed interface.
- If the managed device is a device cluster, you can only select from a list SFRP IP addresses.
- If you selected a managed device **not** managed by the Defense Center, specify an IP address for the endpoint.

Protected Networks

Specify the networks in your deployment that are encrypted. Enter a subnet with CIDR block for each network.

Note that VPN endpoints cannot have the same IP address and that protected networks in a VPN endpoint pair cannot overlap. If a list of protected networks for an endpoint contains one or more IPv4 or IPv6 entry, the other endpoint's protected network must have at least one entry of the same type (i.e., IPv4 or IPv6). If it does not, then the other endpoint's IP address must be of the same type and must not overlap with the entries in the protected network. (Use /32 CIDR address blocks for IPv4 and /128 CIDR address blocks for IPv6). If both of these checks fail, the endpoint pair is invalid.

Internal IP

Select the check box if the endpoint resides behind a firewall with network address translation.

Public IP

If you selected **Internal IP**, specify a public IP address for the firewall. If the endpoint is a responder, you must specify this value.

Public IKE Port

If you selected **Internal IP**, specify a single numerical value from 1 to 65535 for the UDP port on the firewall that is being port-forwarded to the internal endpoint. If the endpoint is a responder and the port on the firewall being forwarded is not 500 or 4500, you must specify this value.

TIP! To edit an existing star deployment, click the edit icon (🔧) next to the deployment. You cannot edit the deployment type after you initially save the deployment. To change the deployment type, you must delete the deployment and create a new one. Two users should **not** edit the same deployment simultaneously; however, note that the web interface does not prevent simultaneous editing.

To configure a star deployment:

ACCESS: Admin/Network Admin

1. Select **Devices > VPN**.
The VPN page appears
2. Click **Add**.
The Create New VPN Deployment pop-up window appears.
3. Give the deployment a unique **Name**.
You can use all printable characters, including spaces and special characters.

4. Click **Star** to specify the **Type**.
5. Give the deployment a unique **Pre-shared Key**.
6. Next to **Hub Node**, click the add icon (⊕).
The Add Hub Node pop-up window appears.

Add Hub Node ? ×

Device:

Virtual Router:

Interface:

IP Address:

Protected Networks ⊕

Network

Advanced:

Internal IP: ☐

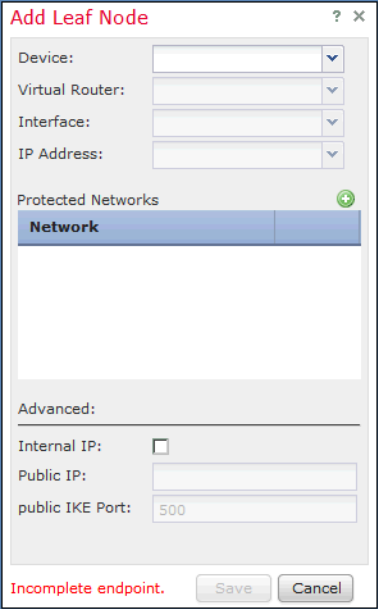
Public IP:

public IKE Port:

Incomplete endpoint.

7. Configure the VPN deployment, as described earlier in this section.
8. Next to **Protected Networks**, click the add icon (⊕).
The Add Network pop-up window appears.
9. Type an IP address for the protected network.
10. Click **OK**.
The protected network is added.
11. Click **Save**.
The hub node is added to your deployment and the Create New VPN Deployment pop-up window appears again.

12. Next to **Leaf Nodes**, click the add icon (⊕).
The Add Leaf Node pop-up window appears.

The image shows a pop-up window titled "Add Leaf Node" with a question mark icon and a close button (X). The window contains several input fields: "Device:" with a dropdown arrow, "Virtual Router:" with a dropdown arrow, "Interface:" with a dropdown arrow, and "IP Address:" with a dropdown arrow. Below these is a section titled "Protected Networks" with a green plus icon and a table with one header row labeled "Network" and one empty data row. At the bottom is an "Advanced:" section with a checkbox for "Internal IP:", a text field for "Public IP:", and a text field for "public IKE Port:" containing the value "500". At the very bottom, there is a red error message "Incomplete endpoint." and two buttons: "Save" and "Cancel".

13. Repeat step 7 through step 10 to complete the leaf node, which has the same options as the hub node.
14. Click **Save**.
The leaf node is added to your deployment and the Create New VPN Deployment pop-up window appears again.
15. Click **Save** to finish configuring your deployment and the VPN page appears again.
Note that you must apply the deployment for it to take effect; see [Applying a VPN Deployment](#) on page 351.

Configuring Mesh VPN Deployments

LICENSE: VPN

SUPPORTED DEVICES: Series 3

When configuring a mesh VPN deployment, you define a group of VPNs to link any two points for a given set of endpoints. For more information, see [Understanding Mesh VPN Deployments](#) on page 335.

The following list describes the options you can specify in your deployment.

Name

Give the deployment a unique name.

Type

Click **Mesh** to specify that you are configuring a mesh deployment.

Pre-shared Key

Define a unique pre-shared key for authentication.

Device

You can select a managed device, including a device stack or cluster, as an endpoint for your deployment. For Sourcefire managed devices not managed by the Defense Center you are using, select **Other** and then specify an IP address for the endpoint.

Virtual Router

If you selected a managed device as your endpoint, select a virtual router that is currently applied to the selected device. You cannot select the same virtual router for more than one endpoint.

Interface

If you selected a managed device as your endpoint, select a routed interface that is assigned to the selected virtual router.

IP Address

- If you selected a managed device as an endpoint, select an IP address that is assigned to the selected routed interface.
- If the managed device is a device cluster, you can only select from a list SFRP IP addresses.
- If you selected a managed device **not** managed by the Defense Center, specify an IP address for the endpoint.

Protected Networks

Specify the networks in your deployment that are encrypted. Enter a subnet with CIDR block for each network. IKE version 1 only supports a single protected network.

Note that VPN endpoints cannot have the same IP address and that protected networks in a VPN endpoint pair cannot overlap. If a list of protected networks for an endpoint contains one or more IPv4 or IPv6 entry, the other endpoint's protected network must have at least one entry of the same type (i.e., IPv4 or IPv6). If it does not, then the other endpoint's IP address must be of the same type and must not overlap with the entries in the protected network. (Use /32 CIDR address blocks for IPv4 and /128 CIDR address blocks for IPv6). If both of these checks fail, the endpoint pair is invalid.

Internal IP

Select the check box if the endpoint resides behind a firewall with network address translation.

Public IP

If you selected **Internal IP**, specify a public IP address for the firewall. If the endpoint is a responder, you must specify this value.

Public IKE Port

If you selected **Internal IP**, specify a single numerical value from 1 to 65535 for the UDP port on the firewall that is being port-forwarded to the internal endpoint. If the endpoint is a responder and the port on the firewall being forwarded is not 500 or 4500, you must specify this value.

TIP! To edit an existing mesh deployment, click the edit icon (✎) next to the deployment. You cannot edit the deployment type after you initially save the deployment. To change the deployment type, you must delete the deployment and create a new one. Two users should **not** edit the same deployment simultaneously; however, note that the web interface does not prevent simultaneous editing.

To configure a mesh VPN deployment:

ACCESS: Admin/Network Admin

1. Select **Devices > VPN**.

The VPN page appears

2. Click **Add**.

The Create New VPN Deployment pop-up window appears.

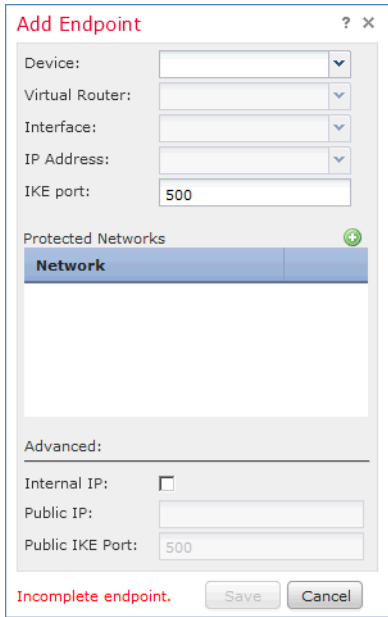
3. Give the deployment a unique **Name**.

You can use all printable characters, including spaces and special characters.

4. Click **Mesh** to specify the **Type**.

5. Give the deployment a unique **Pre-shared Key**.

6. Next to **Nodes**, click the add icon ().
The Add Endpoint pop-up window appears.



The Add Endpoint pop-up window is titled "Add Endpoint" and contains the following fields and sections:

- Device:** A dropdown menu.
- Virtual Router:** A dropdown menu.
- Interface:** A dropdown menu.
- IP Address:** A dropdown menu.
- IKE port:** A text input field with the value "500".
- Protected Networks:** A section with a green plus icon and a table with one header row labeled "Network".
- Advanced:** A section with the following fields:
 - Internal IP:** A checkbox.
 - Public IP:** A text input field.
 - Public IKE Port:** A text input field with the value "500".
- Buttons:** "Save" and "Cancel" buttons.
- Message:** "Incomplete endpoint." in red text.

7. Configure the VPN deployment, as described earlier in this section.
8. Next to **Protected Networks**, click the add icon ().
The Add Network pop-up window appears.
9. Type a CIDR block for the protected network.
10. Click **OK**.
The protected network is added.
11. Click **Save**.
The endpoint is added to your deployment and the Create New VPN Deployment pop-up window appears again.
12. Repeat step 6 through step 11 to add more endpoints.
13. Click **Save** to complete your deployment and the VPN page appears again.
Note that you must apply the deployment for it to take effect; see [Applying a VPN Deployment](#) on page 351.

Configuring Advanced VPN Deployment Settings

LICENSE: VPN

SUPPORTED DEVICES: Series 3

VPN deployments contain some common settings that can be shared among the VPNs in a deployment. Each VPN can use the default settings or you can override

the default settings. Advanced settings typically require little or no modification and are not common to every deployment.

The following list describes the advanced options you can specify in your deployment.

Other Algorithm Allowed

Select the check box to enable auto negotiation to an algorithm not listed in the Algorithm list, but proposed by the remote peer.

Algorithm

Specify the phase one and phase two algorithm proposals to secure data in your deployment. Select **Cipher**, **Hash**, and Diffie-Hellman (**DH**) group authentication messages for both phases.

IKE Life Time

Specify a numerical value and select a time unit for the maximum IKE SA renegotiation interval. You can specify a minimum of 15 minutes and a maximum of 30 days.

IKE v2

Select the check box to specify that the system uses IKE version 2. This version supports the star deployment and multiple protected networks.

Life Time

Specify a numerical value and select a time unit for the maximum SA renegotiation interval. You can specify a minimum of 5 minutes and a maximum of 24 hours.

Life Packets

Specify the number of packets that can be transmitted over an IPsec SA before it expires. You can use any integer between 0 and 18446744073709551615.

Life Bytes

Specify the number of bytes that can be transmitted over an IPsec SA before it expires. You can use any integer between 0 and 18446744073709551615.

AH

Select the check box to specify that the system uses the authentication header security protocol for the data to be protected. Clear the check box to use encryption service payload (ESP) protocol. See [Understanding IPSec](#) on page 333 for guidance on when to use each protocol.

To configure advanced VPN deployment settings:

ACCESS: Admin/Network Admin

1. Select **Devices > VPN**.
The VPN page appears.
2. Click **Add**.
The Create New VPN Deployment pop-up window appears.
3. Click the **Advanced** tab.

The screenshot shows the 'Create New VPN Deployment' dialog box with the 'Advanced' tab selected. The dialog is divided into two main sections: 'IKE Algorithm Proposal' and 'IPSec Algorithm Proposal'. Each section has a table of algorithms and a set of settings.

IKE Algorithm Proposal:

Other Algorithm Allowed: ☐

Cipher	Hash	DH	
aes128	sha1	modp2048	
3des	sha1	modp1536	

IKE Settings:

IKE Life Time: 3 hours

IKE v2: ☒

IPSec Algorithm Proposal:

Other Algorithm Allowed: ☐

Cipher	Hash	DH	
aes128	sha1		
3des	sha1		

IPSec Settings:

Life Time: 1 hours

Life Packets: 0

Life Bytes: 0

AH: ☐

Please enter a deployment name.

Save Cancel

4. Configure the advanced settings, as described earlier in this section.
5. Next to **Algorithms**, click the add icon ().
The Add IKE Algorithm Proposal pop-up window appears.
6. Select **Cipher**, **Hash**, and Diffie-Hellman (**DH**) group authentication messages for both phases.
7. Click **OK**.
The IKE algorithm proposal is added.

8. Click **Save**.

Your changes are saved and the VPN page appears.

Note that you must apply the deployment for it to take effect; see [Applying a VPN Deployment](#) on page 351.

Applying a VPN Deployment

LICENSE: VPN

SUPPORTED DEVICES: Series 3

After configuring or making any changes to a VPN deployment, you must apply the deployment to one or more devices to implement the settings you designated for the deployment.

To apply a VPN deployment:

ACCESS: Admin/Network Admin

1. Select **Devices > VPN**.

The VPN page appears.

2. Click the apply icon (✓) next to the VPN deployment that you want to apply.

3. When prompted, click **Yes**.

The VPN deployment is applied.

TIP! Optionally, from the Apply VPN deployment dialog box, click **View Changes**. The VPN Comparison View page appears in a new browser window. For more information, see [Using the VPN Deployment Comparison View](#) on page 355.

4. Click **OK**.

You are returned to the VPN page.

Viewing VPN Deployment Status

LICENSE: VPN

SUPPORTED DEVICES: Series 3

After you configure a VPN deployment, you can view the status of your configured VPN tunnels. The VPN page displays a status icon for each applied VPN deployment:

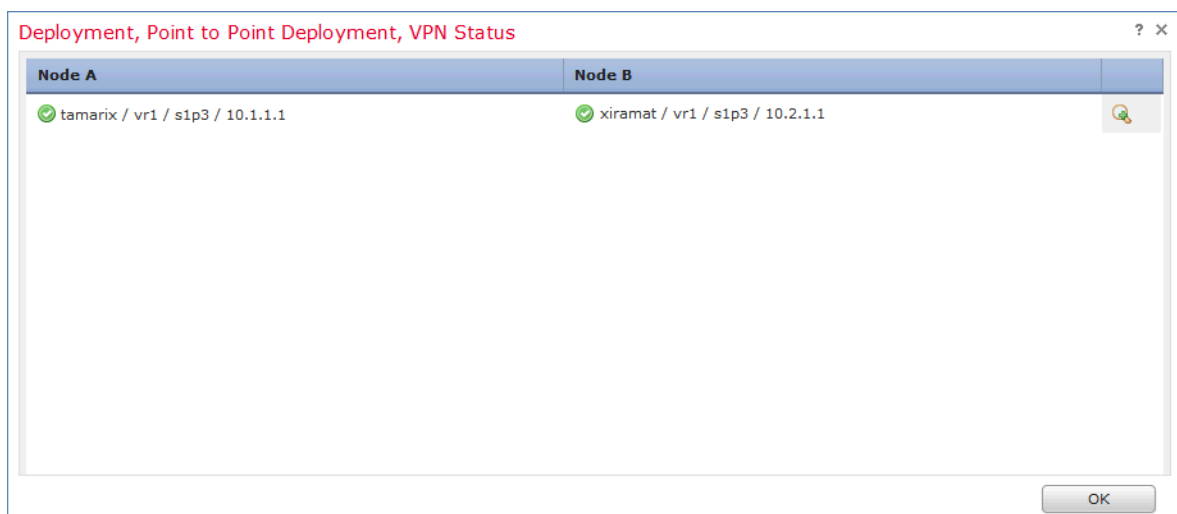
- The (✓) icon designates that all VPN endpoints are up.
- The (⚠) icon designates that all VPN endpoints are down.
- The (⚠) icon designates that some endpoints are up, while others are down.

You can click a status icon to view the deployment status along with basic information about the endpoints in the deployment, such as endpoint name and IP address. The VPN status updates every minute or when a status change occurs, such as an endpoint going down or coming up.

To view VPN status:

ACCESS: Admin/Network Admin

1. Select **Devices > VPN**.
The VPN page appears.
2. Click the VPN status icon next to the deployment where you want to view the status.
The VPN Status pop-up window appears.



3. Click **OK** to return to the VPN page.

Viewing VPN Statistics and Logs

LICENSE: VPN

SUPPORTED DEVICES: Series 3

After you configure a VPN deployment, you can view statistics about the data traversing your configured VPN tunnels. In addition, you can view the latest VPN system and IKE logs for each endpoint.

The system displays the following statistics.

Endpoint

The device path to the routed interface and IP address designated as the VPN endpoint.

Status

Whether the VPN connection is up or down.

Protocol

The protocol used for encryption, either ESP or AH.

Packets Received

The number of packets per interface the VPN tunnel receives during an IPsec SA negotiation.

Packets Forwarded

The number of packets per interface the VPN tunnel transmits during an IPsec SA negotiation.

Bytes Received

The number of bytes per interface the VPN tunnel receives during an IPsec SA negotiation.

Bytes Forwarded

The number of bytes per interface the VPN tunnel transmits during an IPsec SA negotiation.

Time Created

The date and time the VPN connection was created.

Time Last Used

The last time a user initiated a VPN connection.

NAT Traversal

If Yes is displayed, at least one of the VPN endpoints resides behind a device with network address translation.

IKE State

The state of the IKE SA: connecting, established, deleting, or destroying.

IKE Event

The IKE SA event: reauthentication or rekeying.

IKE Event Time

The time in seconds the next event should occur.

IKE Algorithm

The IKE algorithm being used by the VPN deployment.

IPSec State

The state of the IPSec SA: installing, installed, updating, rekeying, deleting, and destroying.

IPSec Event

Notification of when the IPSec SA event is rekeying.

IPSec Event Time

The time in seconds until the next event should occur.

IPSec Algorithm

IPSec algorithm being used by the VPN deployment.

To view VPN statistics:

ACCESS: Admin/Network Admin

1. Select **Devices > VPN**.

The VPN page appears.

2. Click the VPN status icon next to the deployment where you want to view the VPN statistics.

The VPN Status pop-up window appears.

- Click the view statistics icon (📊).

The VPN Statistics pop-up window appears.

VPN Statistics		
	Node A	Node B
Endpoint:	tamarix/vr2/s3p1/10.1.1.3	xiramat/vr1/s3p1/10.1.1.1
Status:	UP	UP
Protocol:	ESP	ESP
Packets Received:	0	0
Packets Forwarded:	54567610	54431977
Bytes Received:	0	0
Bytes Forwarded:	2147483647	2147483647
Time Created:	Tue May 21 03:05:12 2013	Tue May 21 03:04:59 2013
Time Last Used:	Tue May 21 12:50:26 2013	Tue May 21 12:50:26 2013
NAT Traversal:	No	No
IKE Data		
State:	ESTABLISHED	ESTABLISHED
Event:	reauthentication	reauthentication
Event Time:	720	1020
Algorithm:	AES_CBC_128/HMAC_SHA1_96 /PRF_HMAC_SHA1/MODP_1024	AES_CBC_128/HMAC_SHA1_96 /PRF_HMAC_SHA1/MODP_1024
IPSec Data		
State:	INSTALLED	INSTALLED
Event:	Rekeying	Rekeying
Event Time:	2040	2160
Algorithm:	AES_CBC_128/HMAC_SHA1_96	AES_CBC_128/HMAC_SHA1_96
	View Recent Log	View Recent Log
	Refresh	Close

- Optionally, click **Refresh** to update the VPN statistics.
- Optionally, click **View Recent Log** to view the latest data log for each endpoint.
To view the log for clustered devices and stacked devices, you can select the link for either the active/primary or backup/secondary device.

Using the VPN Deployment Comparison View

LICENSE: VPN

SUPPORTED DEVICES: Series 3

The VPN deployment comparison view allows you to view the changes you have made to a deployment before you apply them. The report displays all differences between the current deployment and the proposed deployment. This gives you an opportunity to discover any potential configuration errors.

The comparison view displays both deployments in a side-by-side format, with each deployment identified by name in the title bar on the left and right sides of the comparison view. The time of last modification and the last user to modify are displayed with the deployment name.

Differences between the two deployments are highlighted:

- Blue indicates that the highlighted setting is different in the two deployments, and the difference is noted in red text.
- Green indicates that the highlighted setting appears in one deployment but not the other.

You can perform any of the actions in the following table.

VPN Deployment Comparison View Actions

To...	You CAN...
navigate individually through changes	click Previous or Next above the title bar. The double-arrow icon (↔) centered between the left and right sides moves, and the Difference number adjusts to identify which difference you are viewing.
generate a deployment comparison report	click Comparison Report . The deployment comparison report creates a PDF document that lists only the differences between the two policies.